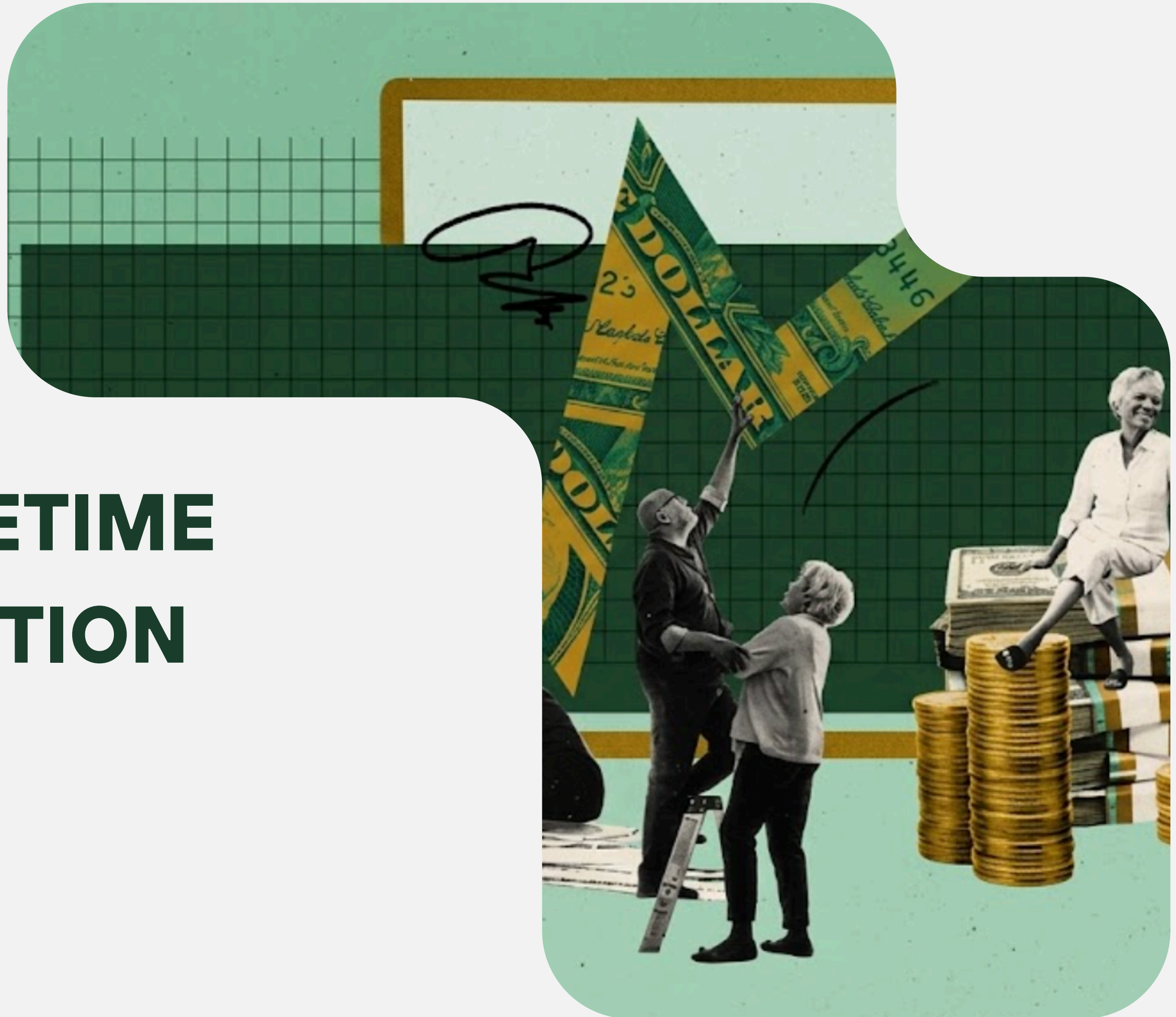


Beatrice Camera, 535313

AI for Communication and
Marketing
Lab Exam – Part 3

CUSTOMER LIFETIME VALUE & RETENTION STRATEGY

From Prediction to Investment





How much is our customer base worth, and which model tells us?

Macro View: Forecasting Future Customer Value

Predicted 3-Month Value of the Repeat-Buyer Base: R\$12,878

BTYD (Probabilistic Modeling) – For the CFO

Calibrated forecast of repeat-buyer value over the next quarter. Forecast of future value among repeat buyers, with interpretable parameters.

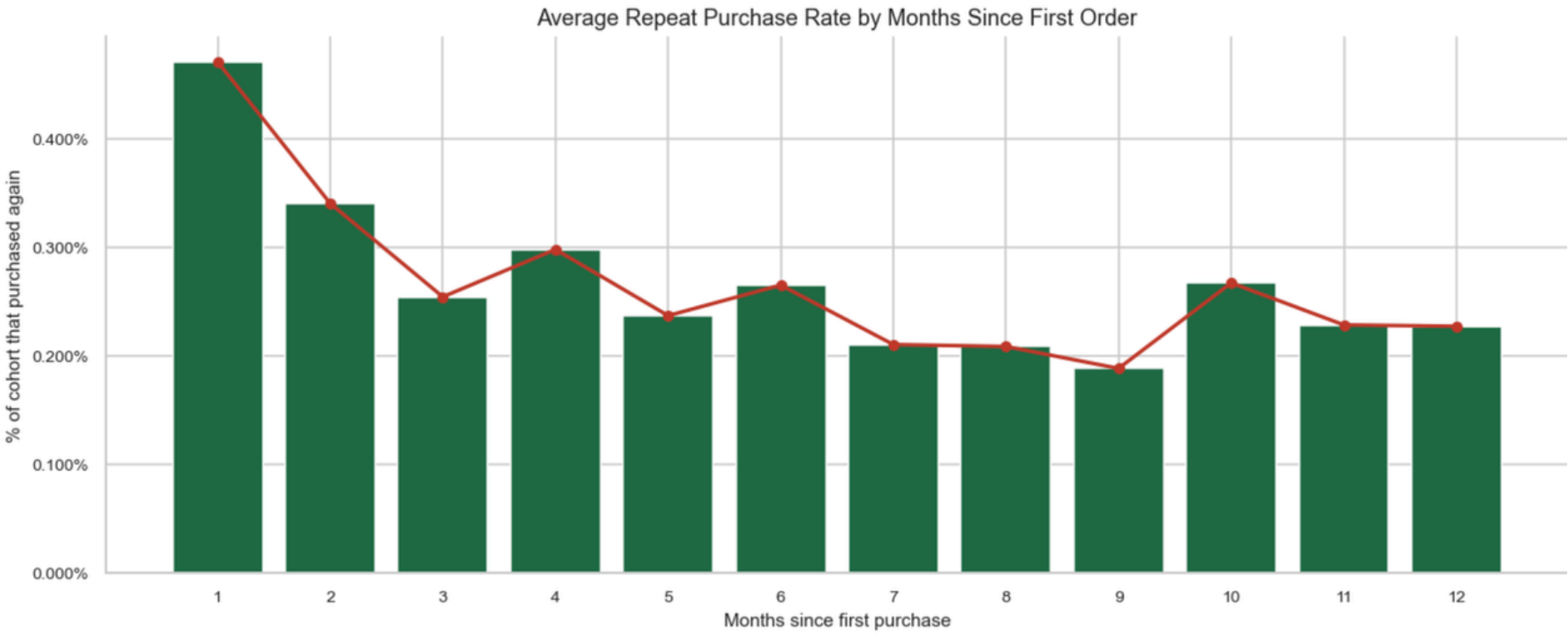
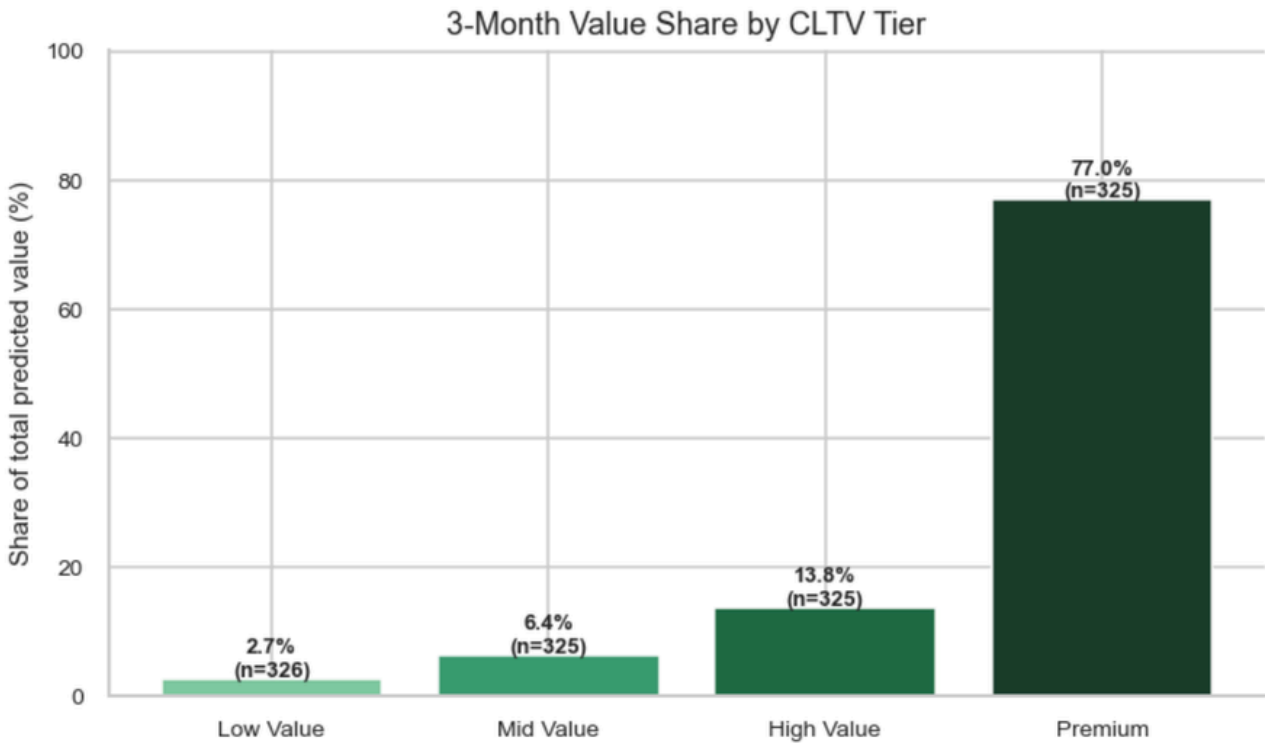
The right instrument for budget planning.

XGBoost (Machine Learning Approach) – For the CRM

Individual CLTV scores across all 86,655 customers.

Sharper rankings, full base coverage.

The right instrument for campaign targeting.



Who to target, and why now?

	WIN-BACK TARGET	RETENTION PRIORITY	GROWTH SEGMENT
WHO	High CLTV + low P(alive)	Premium tier, active	Mid Value, repeat buyers
SIGNAL	53 high-value customers at risk	High predicted value, still engaged	Moderate CLTV, established buyers
ACTION	Past-category recommendations	Loyalty recognition, exclusive access	Category upsell, cross-sell
KPI	Reactivation rate within 60 days	Revenue retention	2nd category purchase rate

Segmentation based on $BTYD\ CLTV \times BG/NBD\ alive\ probability$

